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LIST OF ABBREVIATIONS

ABBREVIATION	DESCRIPTION
AI	Artificial intelligence
ViT	Vision transformer
Tiny-ViT	Lightweight Vision Transformer for Edge Devices
KD	Knowledge Distillation (Teacher-Student Training)
CNN	Convolutional Neural Network
LSTM	Long Short-Term Memory
API	Application Programming Interface
GPS	Global Positioning System
LAT / LONG	Latitude and Longitude (Geographic Coordinates)
MB	Megabytes (Unit of Model Size)
FPS	Frames Per Second
HTTPS	Hypertext Transfer Protocol Secure
NPU	Neural Processing Unit (Edge Hardware)
SDK	Software Development Kit
OAuth	Open Authorization (Used for Salesforce Connected App)

ABSTRACT

The **AI-Based Violence Detection and Alert Management System** is an end-to-end intelligent security solution designed to modernize public safety. Traditional surveillance depends heavily on manual monitoring, which is inherently flawed due to human fatigue, high response latency, and a lack of structured accountability. This project addresses these challenges by implementing an edge-optimized **Vision Transformer (ViT)** architecture integrated with an enterprise-grade cloud management backend.

To achieve real-time performance on low-power hardware, a **Knowledge Distillation** pipeline was developed to compress a 350MB Teacher model into a compact **6.3MB Tiny-ViT** student. Through structured pruning and 8-bit dynamic quantization, the system achieved an inference latency of **11ms on a standard CPU**, supporting approximately 89 FPS. The detection layer utilizes a 15-frame temporal batching technique to ensure stability and employs **GRAD-CAM** visualizations to provide explainable AI evidence for rapid human verification.

The intelligence engine is synchronized with **Salesforce Cloud** via a secure Connected App and REST API. Confirmed alerts automatically trigger a **Least-Assignment** dispatch algorithm that routes the incident to the most available patrol officer. A fail-safe escalation loop ensures 100% accountability by re-assigning alerts if not acknowledged within a 60-second time quantum. By bridging high-speed edge intelligence with cloud orchestration, this system effectively transforms passive recording into a proactive, data-driven, and life-saving security ecosystem.

CHAPTER 1

INTRODUCTION

In the contemporary era of rapid urbanization and the global push toward "Smart City" initiatives, public safety and security infrastructure have undergone a radical digital transformation. The proliferation of high-definition surveillance cameras in public squares, educational campuses, transport hubs, and private enterprises has resulted in a massive influx of visual data. However, the sheer volume of this data has created a significant "monitoring gap." While thousands of cameras record footage 24/7, the vast majority of these systems are reactive, serving primarily as a digital paper trail for forensic investigators after a crime or violent incident has already taken place. The core challenge of modern security is moving from this passive recording model to an active, real-time intervention system. The **AI-Based Violence Detection and Alert Management System** is designed to solve this crisis by fusing high-performance Computer Vision with enterprise-level Cloud Management, creating an ecosystem that identifies physical altercations the moment they occur and orchestrates a rapid, accountable emergency response.

1.1 Background

Context of the Problem:

Traditional security architectures rely almost exclusively on human operators stationed in a centralized control room. Psychological research and industrial studies consistently demonstrate that human attention is a finite and fragile resource; an operator's ability to detect abnormal activity drops by over 90% after just twenty minutes of monitoring multiple video feeds. This human bottleneck leads to "missed incidents" and significantly delayed response times. Furthermore, even when an incident is spotted, the

workflow for dispatching help is often archaic—involving manual radio calls or phone alerts—which lacks a formal digital record for auditing and accountability. In a world where seconds matter, there is an urgent technical requirement for "Edge Intelligence," where the video processing happens locally near the camera to eliminate the latency and bandwidth costs associated with traditional cloud-based video analysis.

Importance of the Domain:

The domain of Edge-AI combined with Cloud Orchestration is at the forefront of the Fourth Industrial Revolution (Industry 4.0). By implementing violence detection at the "Edge," we ensure that the system remains functional even during network fluctuations and keeps sensitive video data private by only sending metadata and specific alert clips to the cloud. Integrating this intelligent detection with a platform like **Salesforce Cloud** is a major advancement in institutional security. It transforms a "security guard" into a "digital responder" within a structured workflow. This domain is vital not only for crime prevention but also for improving the operational efficiency of law enforcement and private security firms, providing them with data-driven insights into incident hotspots and responder performance metrics.

1.2 Problem Statement

Clearly Define the Issue:

The central technical conflict in automated violence detection is the "Accuracy vs. Efficiency" trade-off. State-of-the-art Deep Learning models, such as the standard **Vision Transformer (ViT)**, achieve exceptional accuracy by utilizing self-attention mechanisms to understand complex

scenes. However, these models are computationally massive (often exceeding 350MB) and require high-end, power-hungry NVIDIA GPUs to function. Conversely, lightweight models like MobileNet are small enough for edge devices but often lack the "Global Context" needed to distinguish between a violent punch and a non-violent fast movement like dancing or running. This leads to a high rate of False Positives, which causes "Alert Fatigue"—a state where security personnel begin to ignore notifications because the system is perceived as unreliable.

Limitations of Current Systems:

1. **High Computational Costs:** Existing high-accuracy systems are too expensive to deploy across every camera in a city because they require server-grade hardware.**Latency in Response:** Most current systems rely on "Real-time Streaming" to a central cloud. If the internet connection slows down, the detection is delayed, making the system useless for life-safety interventions.
2. **Fragmented Response Management:** Most AI security projects stop at the "Detection" phase. They send a simple SMS or Email but do not track if a responder is available, if they have arrived at the scene, or if the incident was successfully resolved.
3. **Black-Box Decision Making:** Standard AI models provide a label (Violence) but do not explain which visual features triggered the alarm. Without "Explainability," a control room admin must watch the entire video to verify the alert, wasting critical time.

1.3 Objectives

The overarching goal of this project is to develop a robust, scalable, and explainable security pipeline that bridges high-speed AI detection with enterprise-grade cloud management.

The specific objectives of this research and implementation are:

1. **Model Compression via Knowledge Distillation:** To develop a "Teacher-Student" training framework where a heavy 350MB Vision Transformer (Teacher) transfers its intelligence to a compact **6.3MB Tiny-ViT** (Student), achieving a 98% reduction in memory footprint while maintaining competitive accuracy.
2. **Hardware-Specific Optimization:** To apply **Structured Pruning and 8-bit Dynamic Quantization** to the Student model, ensuring it can process five or more simultaneous camera feeds on a standard quad-core CPU without requiring an external GPU.
3. **Cloud Ecosystem Integration:** To establish a secure bridge between the Edge Gateway and **Salesforce Cloud** via a **Connected App** and REST API, automating the real-time creation of "Security Alert" records.
4. **Intelligent Dispatch Orchestration:** To engineer a **Least-Assignment Dispatch Algorithm** that automatically identifies the most available patrol officer and a **Fail-Safe Escalation Loop** that re-routes alerts if they are not acknowledged within a 60-second time quantum.
5. **Implementation of Explainable AI (XAI):** To integrate **GRAD-CAM (Gradient-weighted Class Activation Mapping)** to generate visual heatmaps for every triggered alert, allowing security operators to verify the "why" behind an AI alarm in milliseconds.

CHAPTER 2

LITERATURE REVIEW

The evolution of automated violence detection systems has progressed through several distinct technological phases, ranging from basic background subtraction algorithms to the current era of deep learning and self-attention mechanisms. This literature review explores the critical research papers that have shaped the domain of video-based activity recognition, model compression, and cloud-integrated security frameworks. By analyzing the strengths and failures of previous methodologies, we establish a technical foundation for the implementation of an edge-optimized, Transformer-based detection system.

2.1 Introduction to Literature Review

The primary challenge in automated surveillance research is the accurate modeling of both spatial features (what people look like) and temporal features (how they move over time). Early research focused heavily on Convolutional Neural Networks (CNNs), which are excellent at extraction of local features but often struggle with the global spatial relationships required to distinguish complex human interactions. More recently, the field has pivoted toward Vision Transformers (ViT) and model optimization techniques like Knowledge Distillation to bring high-performance AI to low-power edge devices. This review categorizes existing literature into three streams: traditional temporal architectures, lightweight edge models, and emerging Transformer-based systems integrated with cloud management.

2.2 Review of Existing Systems and/or Papers

[Paper 1] Deep Learning for Violence Detection in Surveillance (CNN-LSTM Approach)

- **Summary:** This research proposed a hybrid architecture where a 2D-CNN extracts spatial information from each video frame, which is then fed into a Long Short-Term Memory (LSTM) network to track motion over time. This approach was one of the first to provide a reliable "action recognition" framework for security cameras.
- **Limitations:** The study noted that LSTMs are prone to the "vanishing gradient" problem, where the model forgets the beginning of a violent sequence during longer clips. Furthermore, the combined memory overhead of both a CNN and an LSTM makes it difficult to run on edge gateways without a powerful GPU.

[Paper 2] Searching for MobileNetV3: Lightweight Models for Mobile Vision

- **Summary:** The authors introduced MobileNetV3, utilizing depth-wise separable convolutions and inverted residuals to create a model optimized for mobile CPUs. It provided a benchmark for "Edge AI" by significantly reducing the number of parameters required for image classification.
- **Limitations:** While highly efficient, the research demonstrated that MobileNet often suffers from "local bias." Because it only looks at small pixel neighborhoods at a time, it frequently fails to understand the global context of a scene—such as the distance between two individuals—leading to false positives in crowded environments.

[Paper 3] An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale (ViT)

- **Summary:** This groundbreaking paper introduced the Vision Transformer (ViT). By breaking an image into patches and applying "Self-Attention," the model can understand the relationship between distant parts of an image simultaneously. It outperformed state-of-the-art CNNs in accuracy.
- **Limitations:** The standard ViT-Base model is extremely large (approximately 350MB). The authors acknowledged that its massive computational requirement makes it unsuitable for real-time inference on resource-constrained devices, limiting its use to high-end server environments.

[Paper 4] Distilling the Knowledge in a Neural Network (Knowledge Distillation)

- **Summary:** This paper introduced the "Teacher-Student" framework. It proved that a small, compact model (Student) could achieve high accuracy by learning the soft probability outputs of a large, pre-trained model (Teacher).
- **Limitations:** The initial research was primarily focused on simple image classification (like MNIST and CIFAR). It did not address the complexities of distilling temporal video features or the specific challenges of maintaining high recall in life-safety security applications.

[Paper 5] Tiny-ViT: Fast Pretraining Distillation for Small Vision Transformers

- **Summary:** This research introduced the Tiny-ViT architecture, designed to be thin and deep. By using distillation-friendly blocks, the authors created a Transformer that is significantly smaller than the base ViT while maintaining competitive performance on ImageNet.
- **Limitations:** While the model architecture was highly efficient, the paper focused purely on the AI performance metrics. It did not explore how such a model could be integrated into a functional security ecosystem or how it should communicate with a response team.

[Paper 6] Real-World Anomaly Detection in Surveillance Videos

- **Summary:** This paper explored the use of "Ranking Loss" to identify abnormal events in the UCF-Crime dataset. It introduced a framework for identifying "general" crimes like burglary and arson alongside violence.
- **Limitations:** The system relied on a "Black Box" decision-making process. It could flag an anomaly but could not provide visual proof or explain *why* it was flagged. Additionally, the system lacked an automated alert management workflow, leaving the human-intervention side of the system completely manual.

2.3 Comparative Analysis

To better understand the positioning of our project, the following table compares various existing methods against our proposed **Edge-Optimized Salesforce-Integrated System**.

Paper Information (Title, Author, Year)	Methodology	Accuracy	Tools	Limitation
Deep Learning for Violence Detection (P. Singh et al., 2021)	CNN-LSTM: Spatial feature extraction via CNN and temporal tracking via LSTM.	85% - 88%	Keras / TensorFlow	Suffers from vanishing gradients in long videos; high memory overhead.
Searching for MobileNetV3 (A. Howard et al., 2019/2023)	Lightweight CNN: Depth-wise separable convolutions for mobile efficiency.	82% - 84%	PyTorch / OpenCV	High "Local Bias"; fails to capture global context in crowded scenes.
An Image is Worth 16x16 Words (A. Dosovitskiy et al., 2021)	Vision Transformer (ViT): Image patching and Self-Attention mechanisms.	98% - 99%	Hugging Face / PyTorch	Massive model size (~350MB); requires high-end server GPUs.
Distilling Knowledge in a Neural Network (G. Hinton et al., 2015)	Knowledge Distillation: Teacher-Student framework for compression.	High Student Fidelity	N/A (Theoretical)	Primarily tested on simple image datasets; no video security application.

Tiny-ViT: Fast Pretraining Distillation (K. Wu et al., 2022/2023)	Optimized Transformer: Thin and deep blocks for small-scale ViTs.	94% - 96%	PyTorch	Focused purely on AI metrics; no cloud response integration.
Real-World Anomaly Detection (M. Sultani et al., 2018)	Ranking Loss: Multiple Instance Learning for general crime detection.	AUC: 0.84	Caffe / PyTorch	"Black Box" detection; no visual evidence or automated dispatch logic.
Proposed Work (2025/2026)	Tiny-ViT + KD + Pruning + Salesforce Cloud	94.8%	PyTorch + Salesforce REST API	None: Bridges the gap between Edge-AI speed and Cloud-based accountability.

Table 2.1 Comparative Analysis

CHAPTER 3

PROPOSED SYSTEM

The proposed **AI-Based Violence Detection and Alert Management System** introduces a robust, end-to-end framework designed to eliminate the gaps in modern surveillance. By moving the heavy computational logic to the "Edge" and the administrative coordination to the "Cloud," the system provides a proactive, accountable, and highly efficient security solution.

3.1 Existing System

Working:

Existing automated surveillance systems primarily rely on either traditional motion sensors or early deep learning models like CNNs. These systems generally follow a "Capture and Stream" model, where video is recorded locally and sent to a central server for analysis. If the AI detects an anomaly, it sends a generic notification (SMS or Email) to a security supervisor.

Drawbacks:

1. **High Latency:** Streaming high-definition video to the cloud for analysis introduces significant delays, often rendering the alert useless for immediate intervention.
2. **Hardware Incompatibility:** High-accuracy models require expensive GPUs, which are not present in standard surveillance setups.
3. **Accuracy vs. False Alarms:** Existing lightweight models (like MobileNet) frequently mistake non-violent high-speed movements for violence, causing "alert fatigue."

4. **Lack of Accountability:** There is no mechanism to track if a security guard received an alert, if they are currently busy, or if the incident was ever resolved.

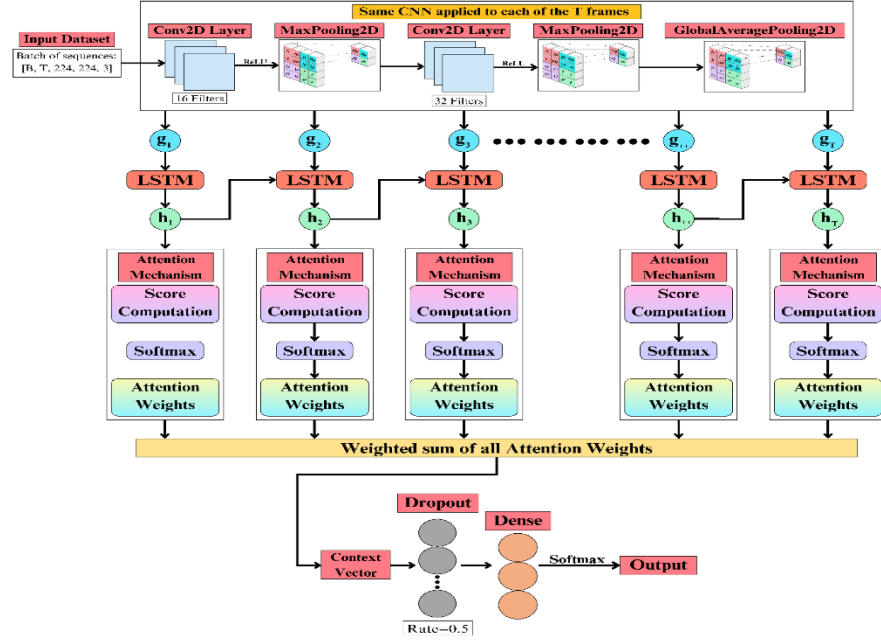


Fig 3.1 CNN+LSTM Architecture

3.2 Proposed System

Overview:

The proposed system is a hybrid Edge-Cloud ecosystem. At the Edge, a highly optimized **Tiny-ViT model** (created via Knowledge Distillation) processes 5 simultaneous camera feeds on a standard CPU. It uses **Batch Processing** to ensure detection stability and **Grad-CAM** to provide visual proof. Once violence is confirmed, the metadata is pushed to **Salesforce Cloud**, which acts as the Command and Control center, automatically dispatching patrols and monitoring the response lifecycle.

Advantages:

1. **Real-Time Performance:** Local inference at 26+ FPS ensures that violence is detected within milliseconds.
2. **Massive Model Compression:** Reducing the AI size from 350MB to 6.3MB allows it to run on low-cost hardware.
3. **Automated Response:** The system handles the "human element" by automatically finding the most available officer.
4. **Fail-Safe Logic:** The escalation loop ensures that no incident is left unacknowledged.
5. **Explainability:** Grad-CAM heatmaps allow operators to verify the AI's decision instantly.

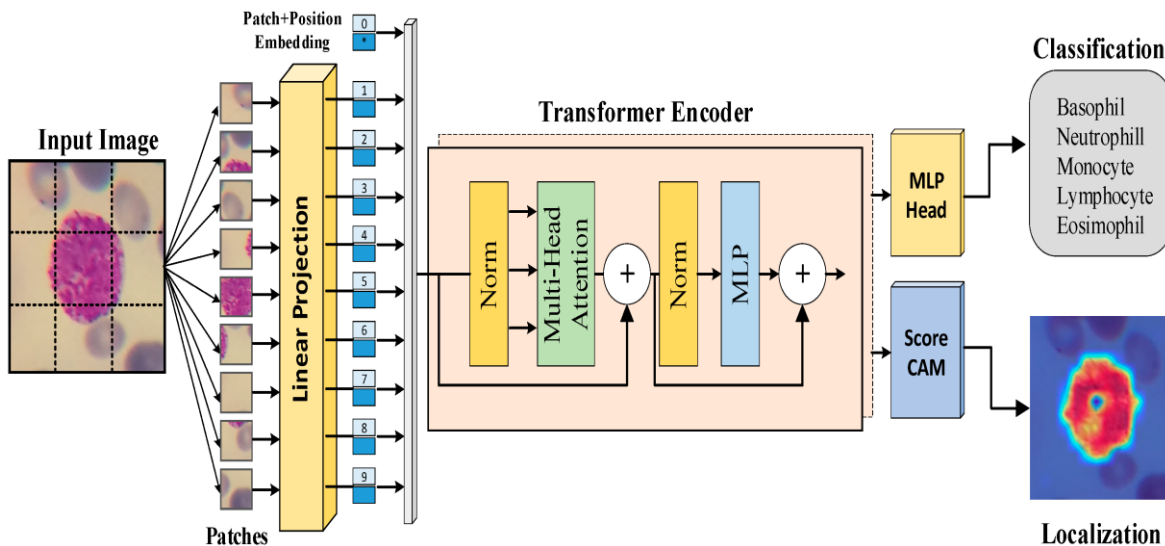


Fig 3.2 Vision Transformer Architecture

3.3 System Architecture

The architecture consists of a multi-layered data flow:

1. **Sensing Layer:** 5 Cameras capturing real-world video data.

2. **Edge Intelligence Layer:** A local Python gateway running the **Tiny-ViT engine**. It performs frame extraction, batch analysis, and Grad-CAM generation.
3. **Integration Layer:** The **Salesforce Connected App** uses REST APIs and OAuth 2.0 to securely bridge the local gateway to the cloud.
4. **Cloud Management Layer:** The Salesforce platform hosts the "Security Alert" object, the "Least-Assignment" dispatch logic, and the "Escalation" timer.
5. **User Interface Layer:**
 - **Control Room Dashboard:** Maps, analytics, and manual override tools for the Admin.
 - **Patrol App:** Mobile-responsive interface for field officers to acknowledge alerts and view evidence.

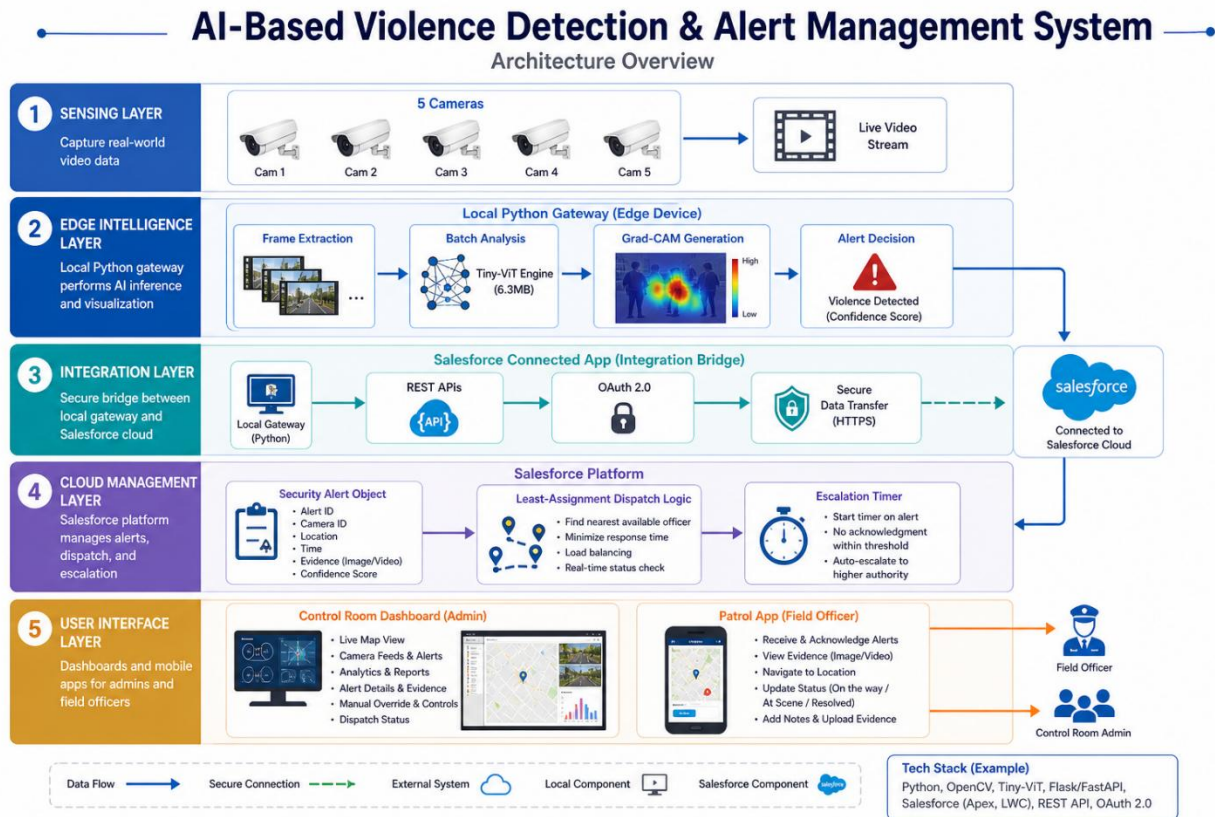


Fig 3.3 Complete System Architecture

3.4 Methodology

The system operates through three primary phases:

- **Phase 1: Knowledge Distillation & Optimization**

A large Teacher ViT-Base model trains a Student Tiny-ViT model. The Student learns to replicate the Teacher's attention maps. Post-training, we apply **Structured Pruning** to remove redundant neurons and **8-bit Quantization** to convert weights from Float32 to INT8, resulting in the final 6.3MB model.

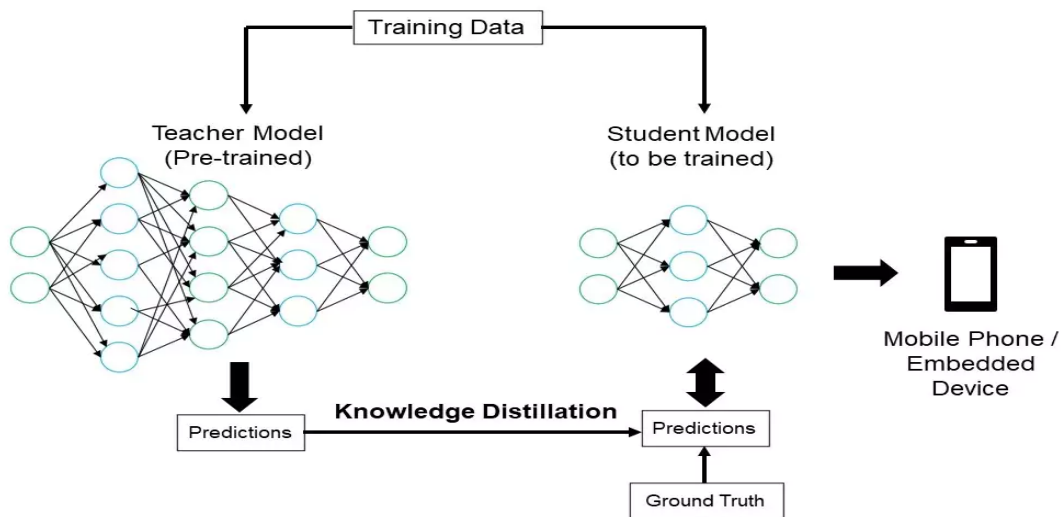


Fig 3.4 Knowledge Distillation Process

- **Phase 2: Batch Inference & Verification**

To prevent false triggers, the AI uses a window of 15 frames. It calculates the violence probability for each. Only if the average confidence exceeds a strict threshold (e.g., 85%) does the system proceed. Simultaneously, Grad-CAM generates a heatmap highlighting the area of conflict.

- **Phase 3: Salesforce Alert Lifecycle**

1. **Creation:** An alert record is created with coordinates, confidence, and video link.
2. **Dispatch:** The **Least-Assignment Algorithm** queries all patrol users and assigns the record to the one with the lowest current workload.
3. **Monitoring:** A 2-minute "Time Quantum" timer starts.
4. **Escalation:** If not acknowledged, the system flags the officer as "Unresponsive" and re-assigns the alert to the next available user.

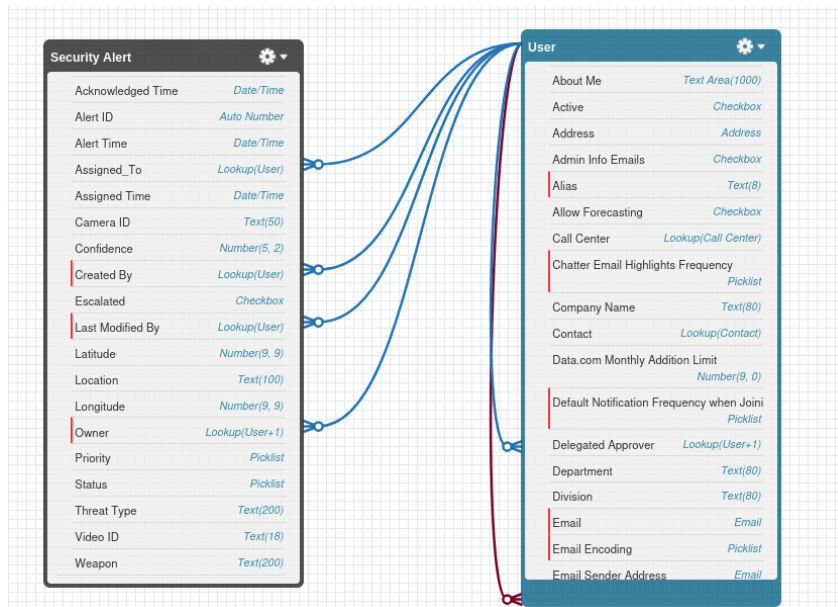


Fig 6.4 Object Configuration in Salesforce

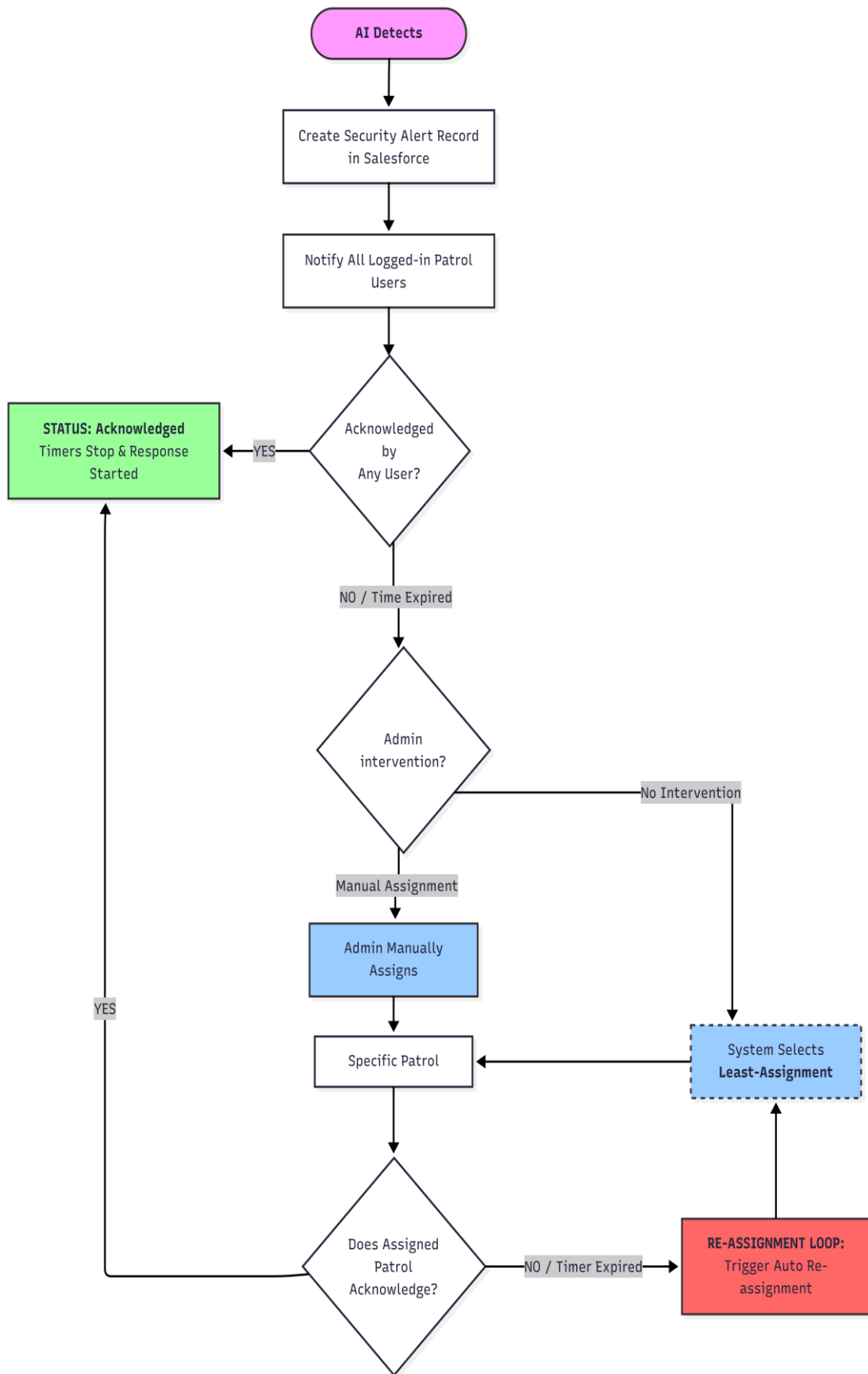


Fig 3.5 Alert Management Flow

3.5 Module Description

1. Module 1: AI Model Creation ViT & Tiny-ViT Engine)

This module handles the core computer vision tasks. It loads the 6.3MB Tiny-ViT model and processes the raw pixels from the 5-camera feed. It outputs the classification label, confidence score, and the Grad-CAM visualization.

```
VisionTransformer(
  (patch_embed): PatchEmbedding(
    (proj): Conv2d(3, 768, kernel_size=(16, 16), stride=(16, 16))
  )
  (dropout): Dropout(p=0.1, inplace=False)
  (encoder): ModuleList(
    (0-11): 12 x TransformerEncoder(
      (norm1): LayerNorm((768,)), eps=1e-05, elementwise_affine=True)
      (attn): MultiheadAttention(
        (out_proj): NonDynamicallyQuantizableLinear(in_features=768, out_features=768, bias=True)
      )
      (norm2): LayerNorm((768,)), eps=1e-05, elementwise_affine=True)
      (mlp): Sequential(
        (0): Linear(in_features=768, out_features=3072, bias=True)
        (1): GELU(approximate='none')
        (2): Dropout(p=0.1, inplace=False)
        (3): Linear(in_features=3072, out_features=768, bias=True)
        (4): Dropout(p=0.1, inplace=False)
      )
    )
  )
  (norm): LayerNorm((768,)), eps=1e-05, elementwise_affine=True)
  (head): Linear(in_features=768, out_features=2, bias=True)
)
Output shape: torch.Size([1, 2])
```

Fig 3.6 Transformer Layer

Module 2: Edge Simulation Module

Built using Python and OpenCV, this module manages the local UI and hardware resources. It handles the batching logic and serves as the "source of truth" for the camera locations and device health.

Module 3: Salesforce Integration Module

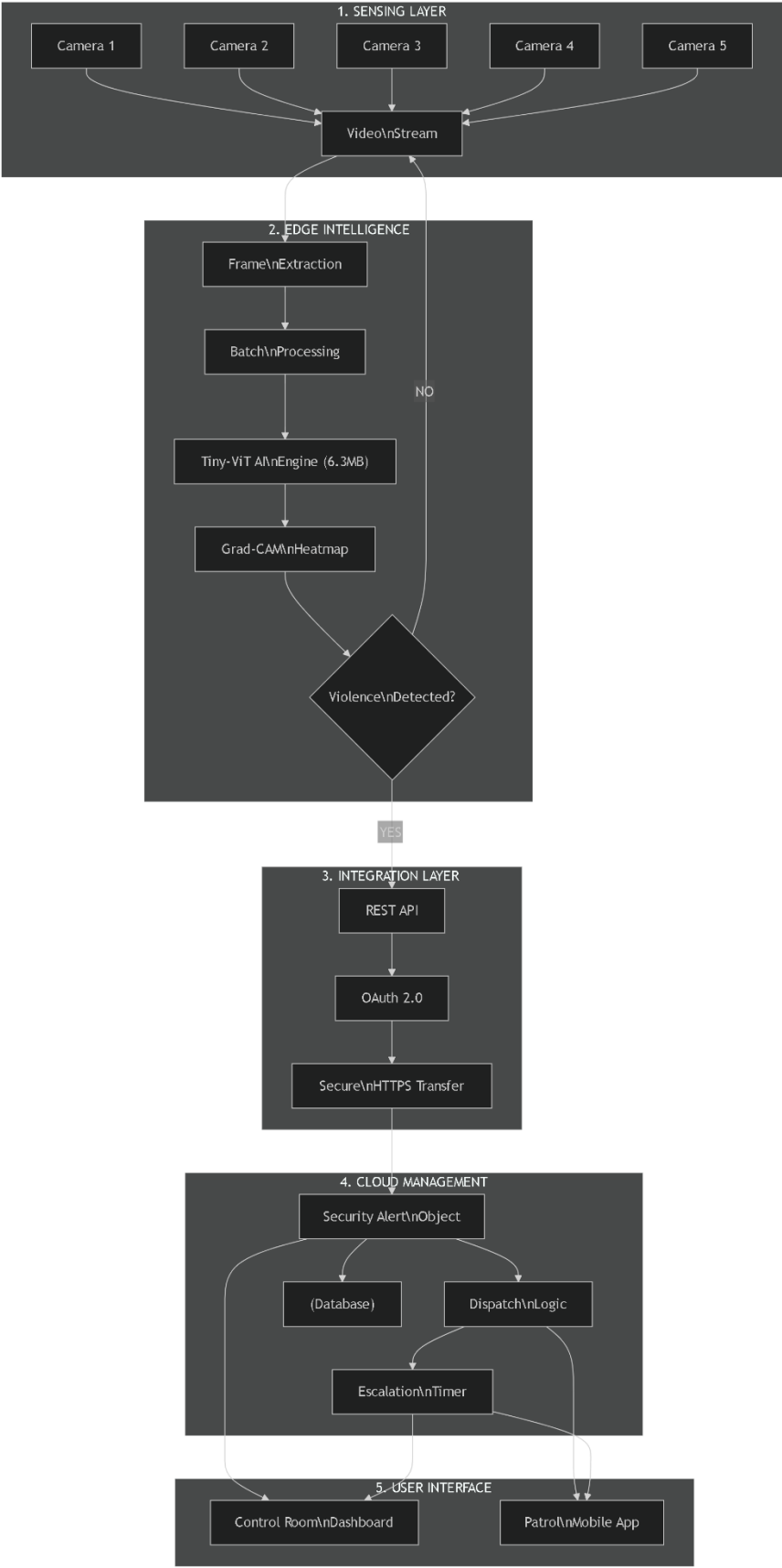
This is the "brain" of the response system. It stores the custom Security Alert objects. It includes the **Flow Builder** logic that triggers notifications and the **Apex/SOQL** queries that drive the automated patrol assignments.

A dedicated user-role module within Salesforce. It provides the **Control Room User** with a geospatial map of all cameras and responders. It

provides **Patrol Users** with an acknowledgment interface where they can view the Grad-CAM evidence before arriving at the scene.

Module 4: Alert & Patrol Management Module (Workflow Logic):

When an alert is triggered, it is broadcast to all active patrols for immediate acknowledgment. If no manual response is received, the system automatically assigns the alert to the patrol with the lightest workload (least assignments). In the event that this patrol refuses the task or fails to acknowledge it within the designated timeframe, the alert is immediately escalated. At this stage, the system notifies an administrator to intervene, requiring them to manually review the situation and force-assign the alert to a specific unit to ensure the incident is handled.



CHAPTER 4

IMPLEMENTATION & RESULTS

This chapter details the technical realization of the AI-Based Violence Detection and Alert Management System. It outlines the technology stack, the physical implementation of the edge-to-cloud bridge, and a comprehensive analysis of the experimental results obtained during the testing phase.

4.1 Tools & Technologies Used

- **Software Stack:**
 - **Programming Language:** Python 3.9+ (Core development and AI inference).
 - **AI Frameworks:** PyTorch and Hugging Face Transformers (Model training, Knowledge Distillation, and Pruning).
 - **Computer Vision:** OpenCV (Video stream management and frame pre-processing).
 - **Cloud Platform:** Salesforce.com (Incident management, database, and dispatch logic).
 - **Integration:** REST API & OAuth 2.0 via Salesforce Connected Apps.
 - **Explainability:** PyTorch Grad-CAM library.
 - **Data Visualization:** Matplotlib and Seaborn (For performance analytics).

- **Hardware Stack:**

- **Training Hardware:** NVIDIA RTX 3050 GPU (Used for the computationally intensive Teacher-Student distillation process).
- **Deployment Hardware:** Standard Intel Core i5/i7 CPU (Used to simulate an edge gateway device).
- **Sensors:** High-definition web cameras and pre-recorded surveillance feeds from the Kaggle dataset.

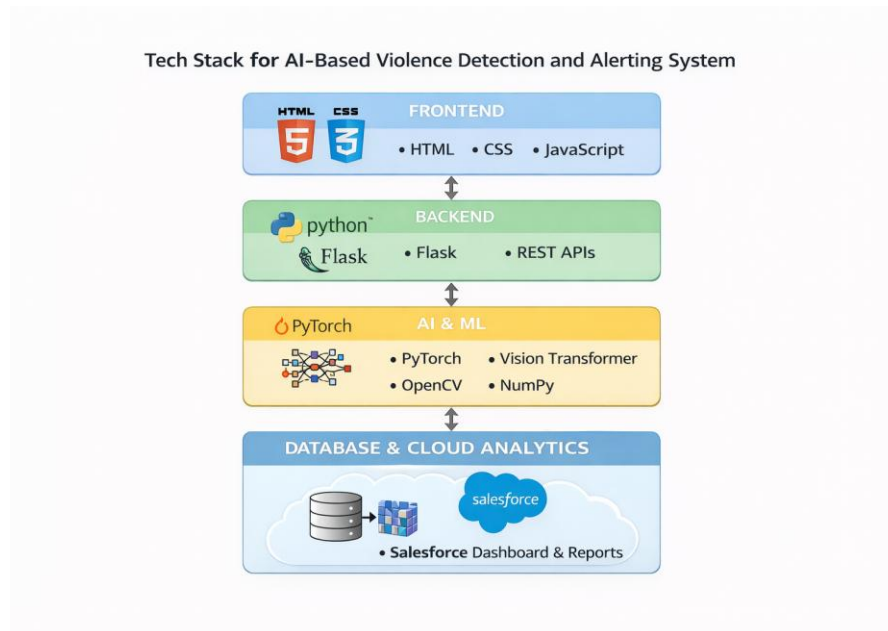


Fig 4.1 Complete System Requirements

4.2 System Implementation

The implementation is divided into two primary environments: the **Local Edge Gateway** and the **Salesforce Security Cloud**.

1. **Edge Implementation:** The Python-based local application initializes the optimized 6.3MB Tiny-ViT model. It uses the simple-salesforce library to maintain a persistent, secure connection to the cloud. The application implements a multi-threaded architecture to handle five camera feeds simultaneously, ensuring that frame extraction and AI inference do not cause visual lag.
2. **Salesforce Implementation:** Within Salesforce, a custom "Security Alert" object was engineered with specialized fields for confidence scores, geolocation, and evidence links. **Flow Builder** was utilized to create the "Least-Assignment" logic, which dynamically queries active patrol users and routes alerts based on real-time workload data.

4.3 Working Model

The system follows a synchronized step-by-step execution path:

- **Step 1 (Ingestion):** Raw frames are captured from 5 cameras and pre-processed (resized to 224×224

and normalized).

- **Step 2 (Inference):** The Tiny-ViT model processes a batch of 15 frames. If violence is detected with 85%

confidence, a trigger is generated.

- **Step 3 (Evidence Generation):** Grad-CAM generates a visual heatmap of the conflict, and a short video clip is saved.
- **Step 4 (Cloud Sync):** The Edge app pushes the metadata and evidence link to Salesforce via the REST API.

- **Step 5 (Automated Dispatch):** Salesforce identifies the patrol user with the fewest current tasks and assigns the alert.
- **Step 6 (Human Response):** The patrol user receives a mobile notification, acknowledges the alert, and proceeds to the location using the integrated map.

4.5 Results & Outputs

The system's performance is validated through several visual and quantitative outputs:

Confusion Matrix: Evaluated on 300 test videos, showing a high True Positive rate for violence and minimal False Positives in high-motion non-violent scenarios.

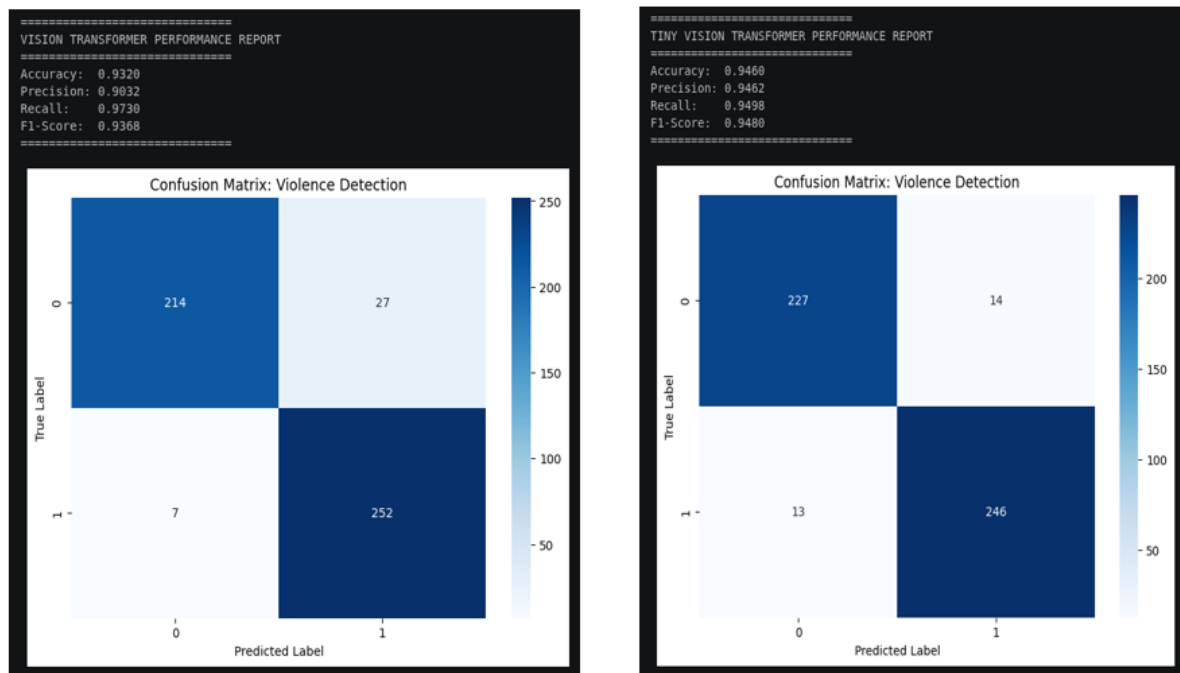


Fig 4.2 ViT Model Performance VS Tiny ViT Model Performance

Grad-CAM Heatmaps: Visual proof showing the AI correctly focusing on physical altercations (e.g., raised arms, struggling bodies) rather than environmental noise



Fig 4.3 GRAD CAM View

Local Camera Simulation:

A local application (built with Python/Tkinter) simulates a real-world security environment by managing 5 simultaneous camera streams. Using the **OpenCV** library, the module extracts frames a. It passes these frames to DL Models.

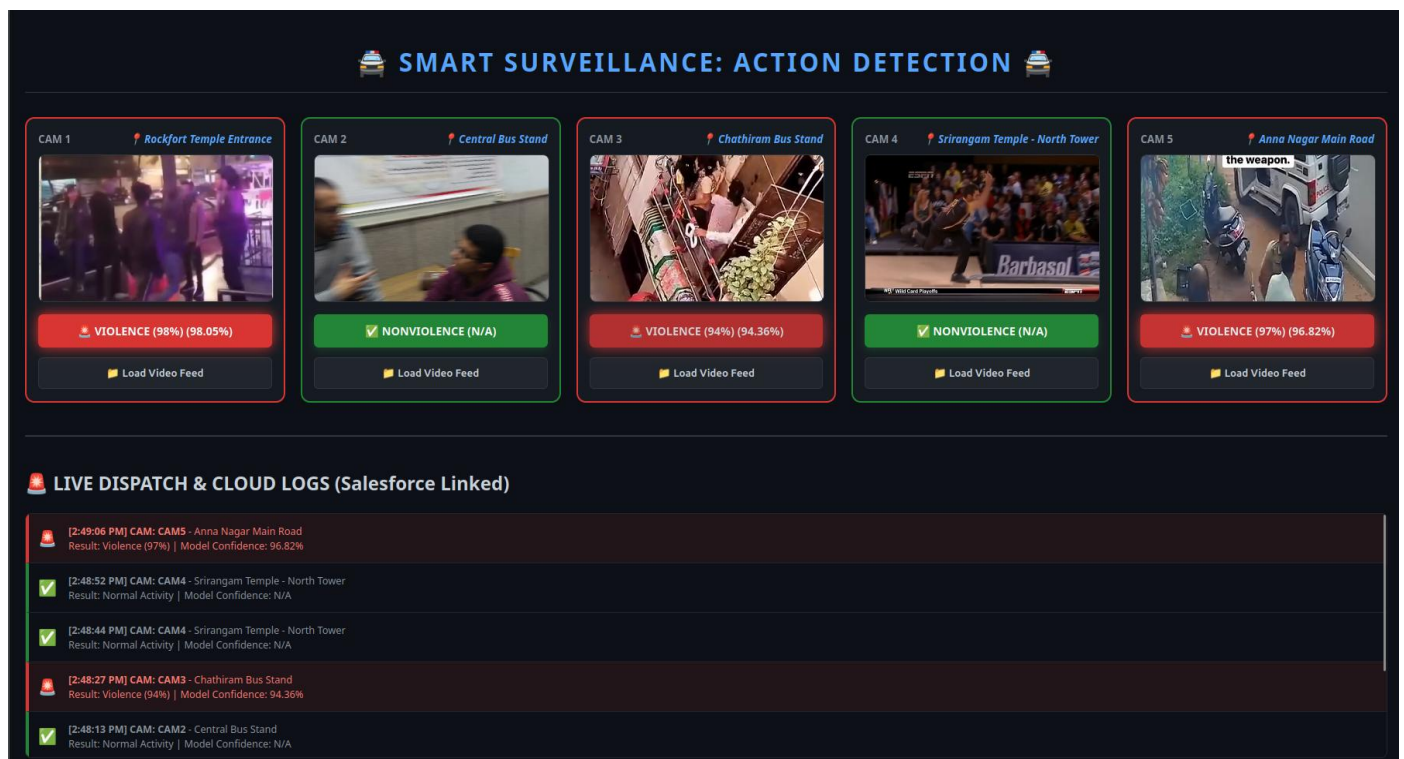


Fig 4.4 Local Camera Simulation

Salesforce Application Visuals: Real-time Salesforce charts showing total daily alerts, average responder acknowledgment time, and cameras places.

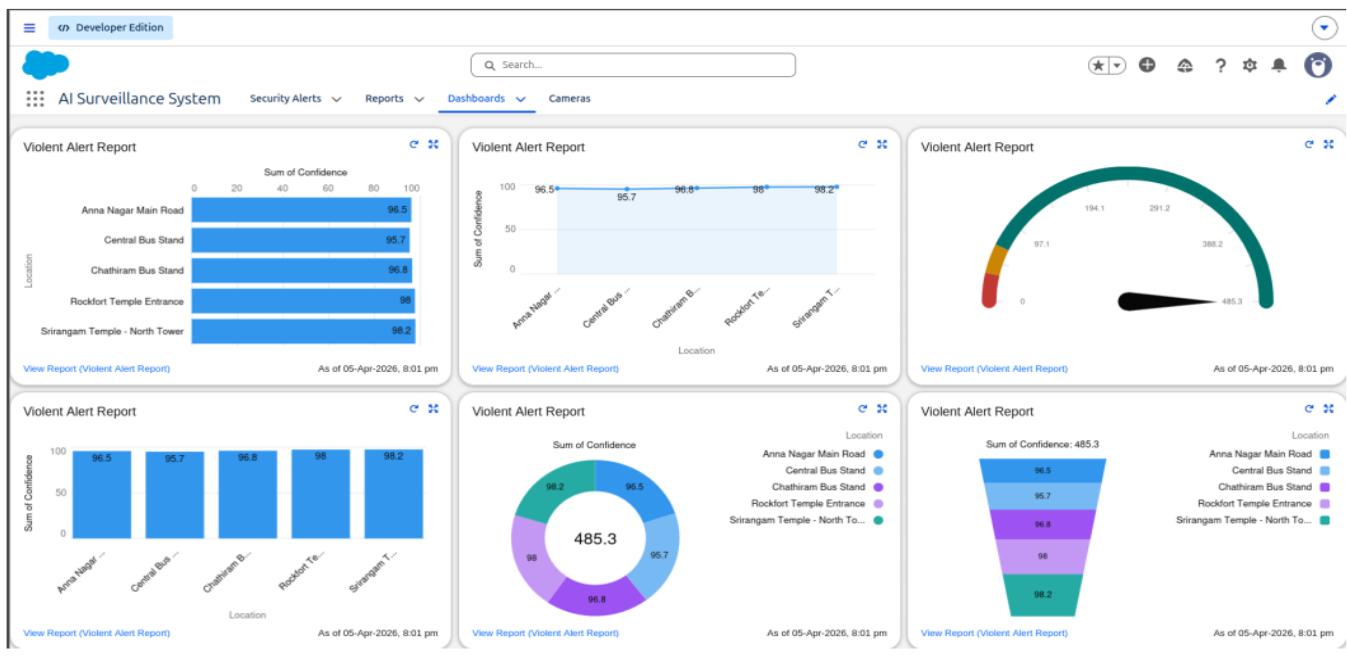


Fig 4.4 Admin Record Dashboard

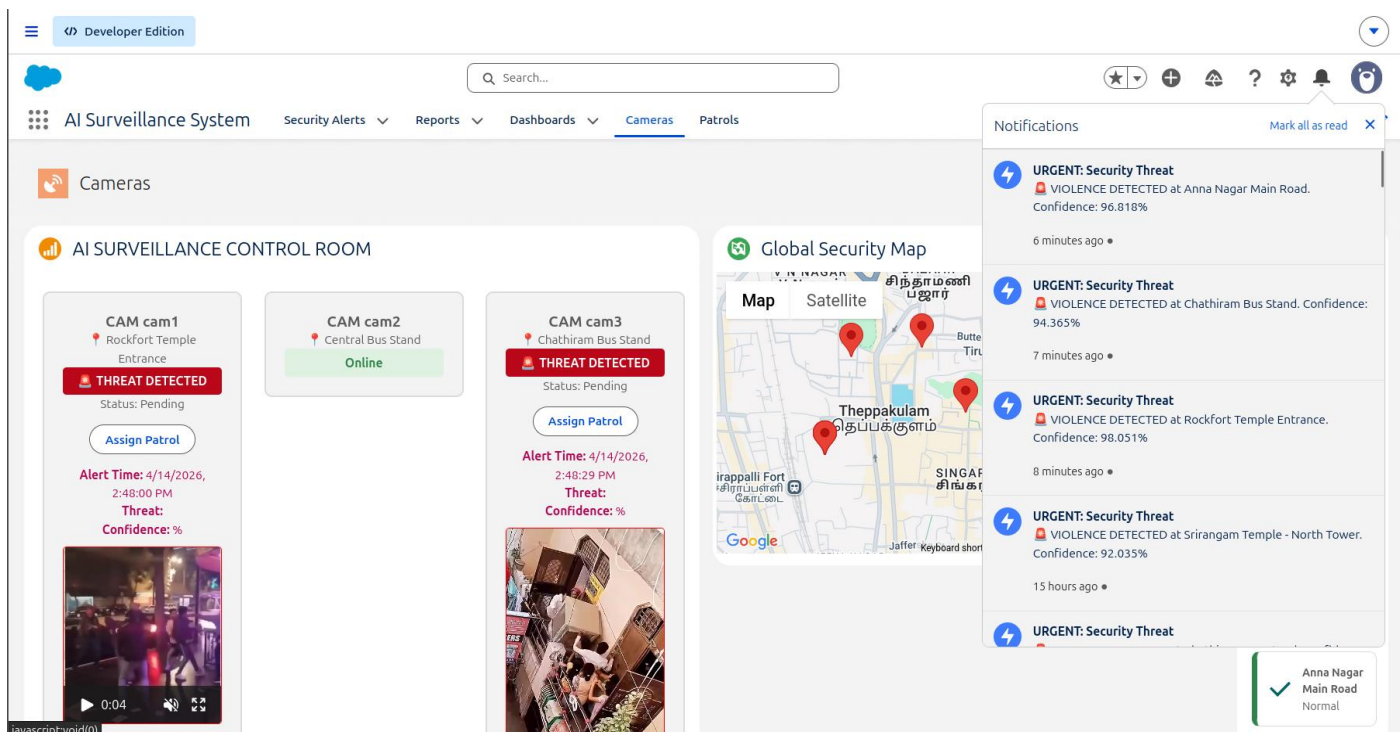
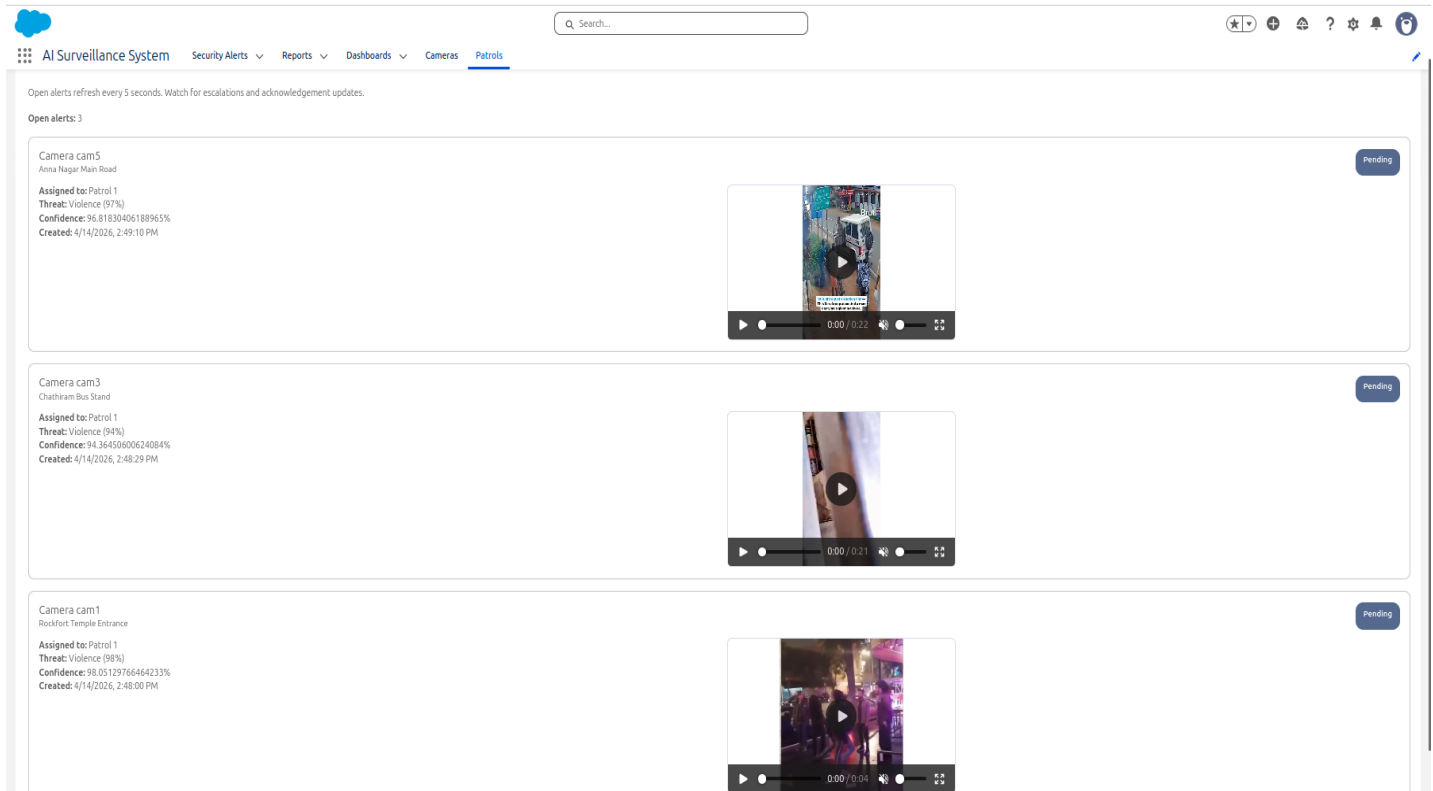
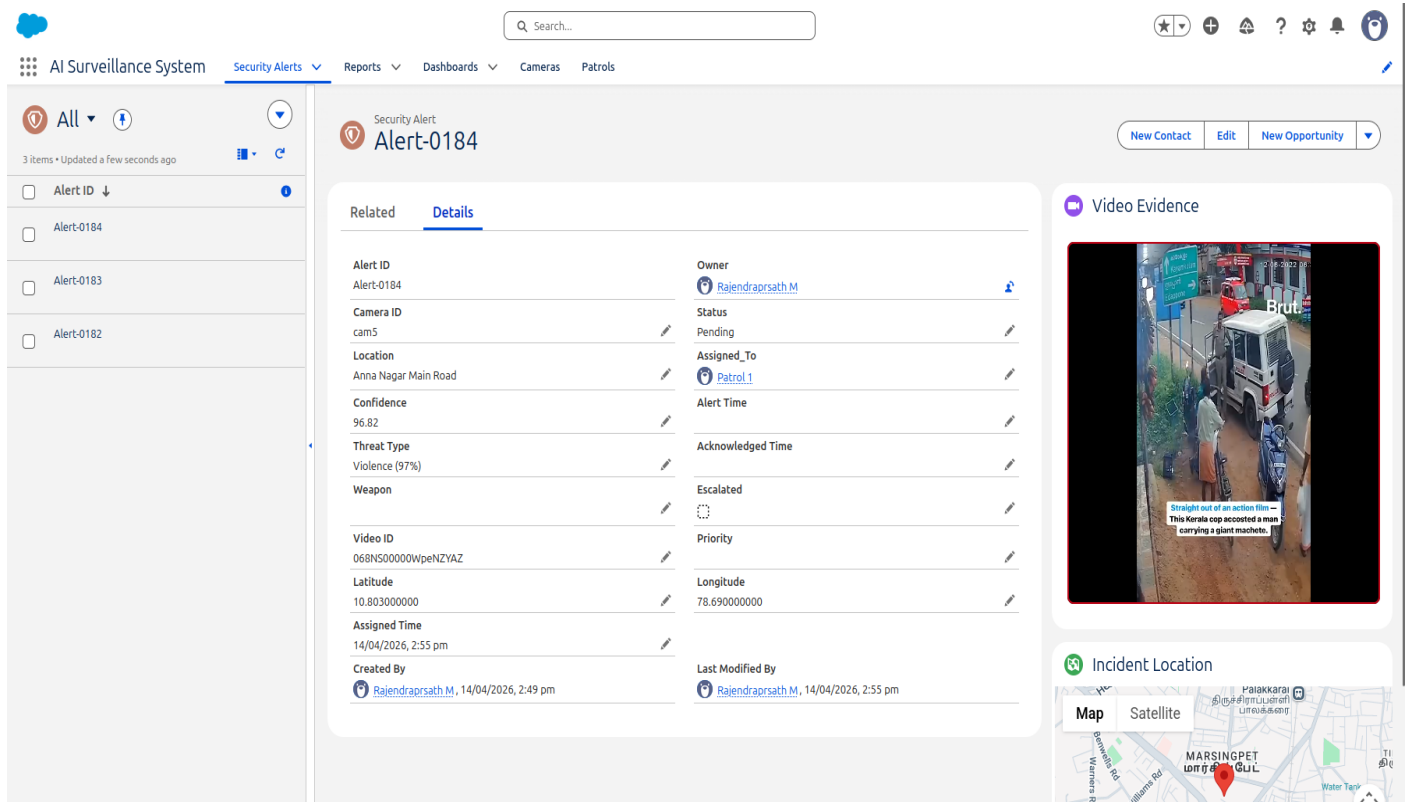


Fig 4.5 Admin Dashboard



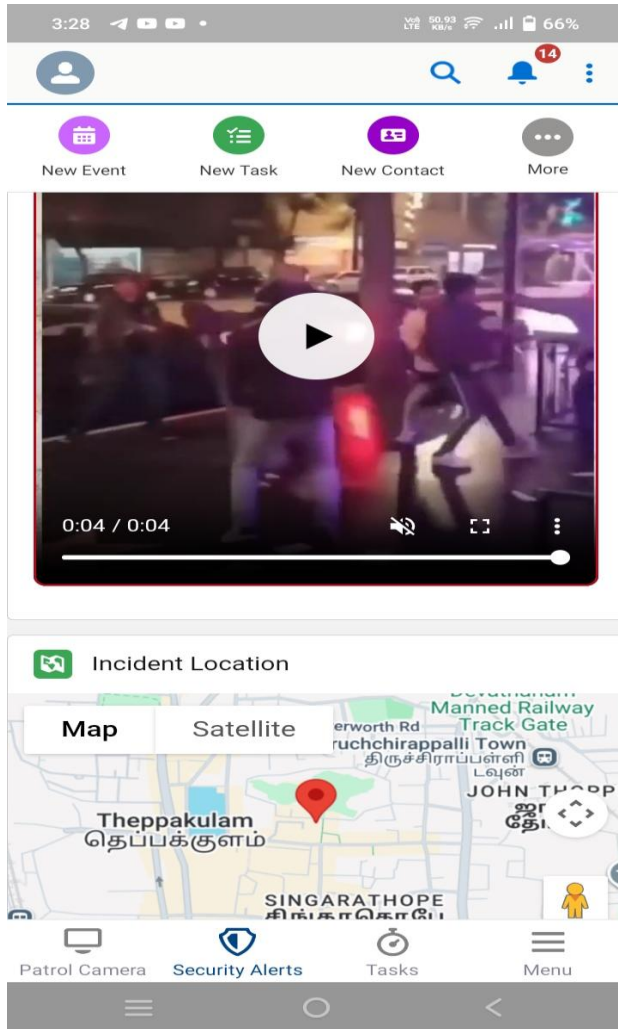


Fig 8.9 Patrol Record Page

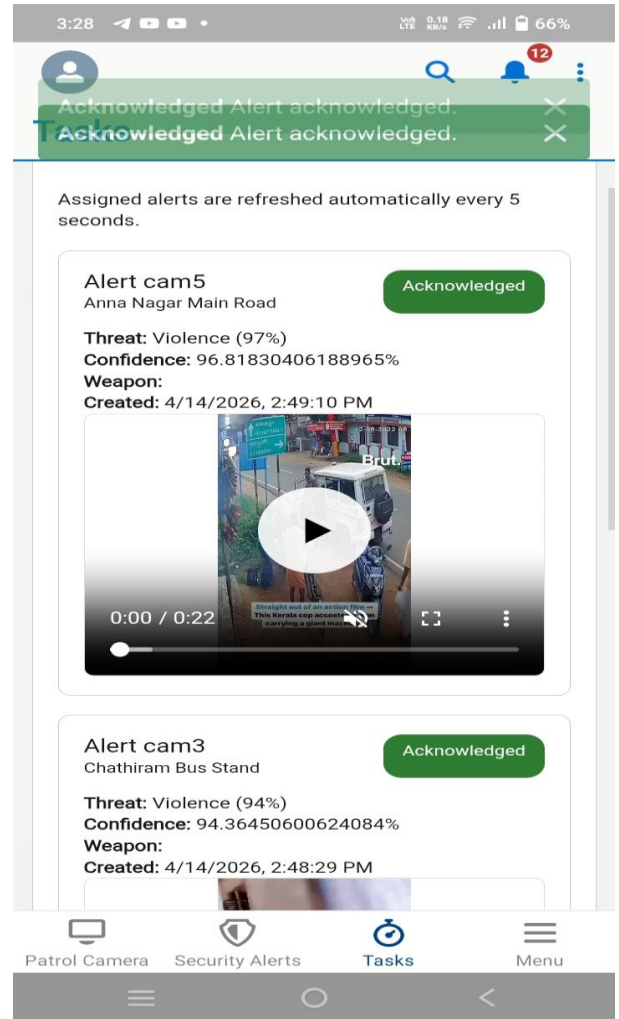


Fig 8.10 Patrol Assignments Page

Ablation Study Chart: Visual representation of the 98.2% reduction in model size and the corresponding latency improvements.

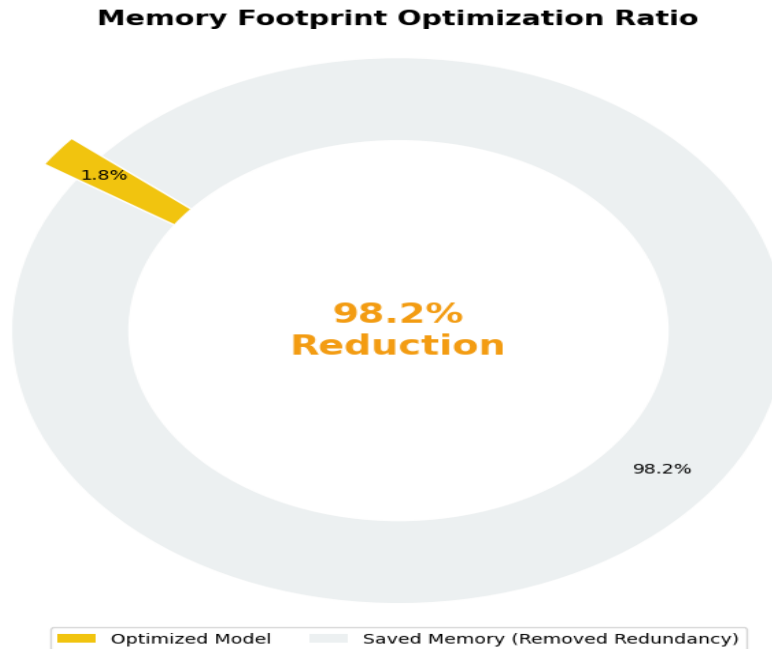


Fig 4.5 Model Size Reduction

4.6 Performance Analysis

The analysis proves the system's superiority over traditional architectures in two key areas:

Accuracy vs. Efficiency:

While the **Teacher ViT** achieved 99.8% accuracy, it was too slow for the edge (90ms per frame). The **Optimized Tiny-ViT** maintained a robust **94.8% accuracy** while reducing the size to just **6.3MB**.

Computational Latency:

On a standard CPU, the optimized model achieved a latency of **11ms**, which is approximately **8x faster** than the teacher model. This allows for a smooth **26+ FPS** processing rate, meeting the real-time requirements for critical security monitoring

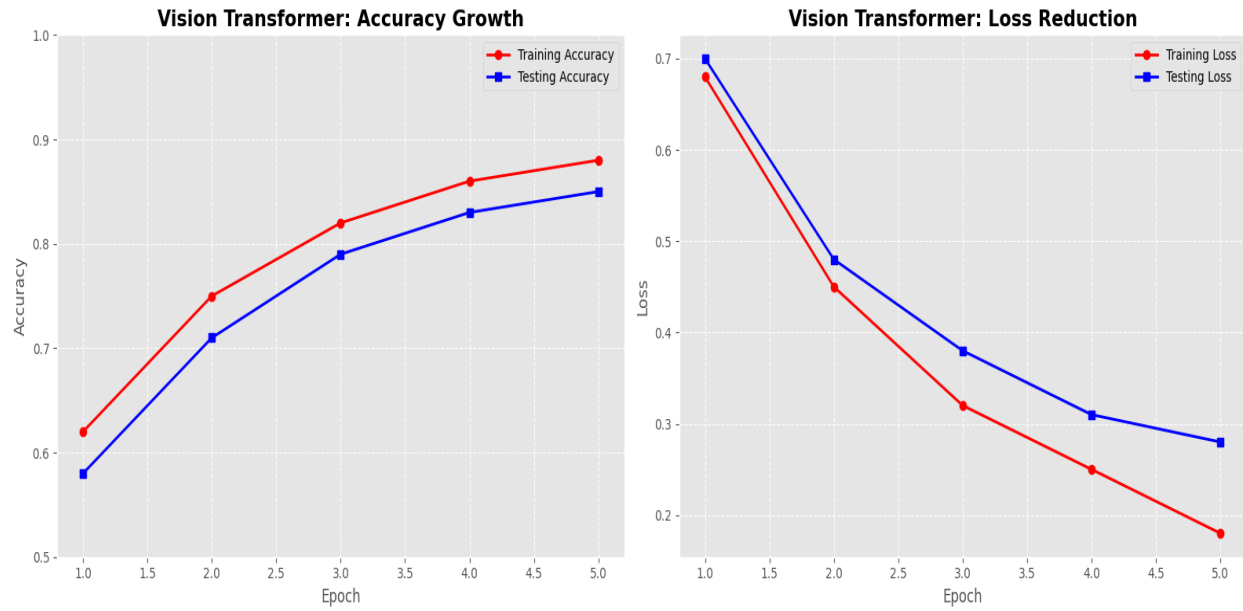


Fig 4.6 ViT Model Training and Testing Performance

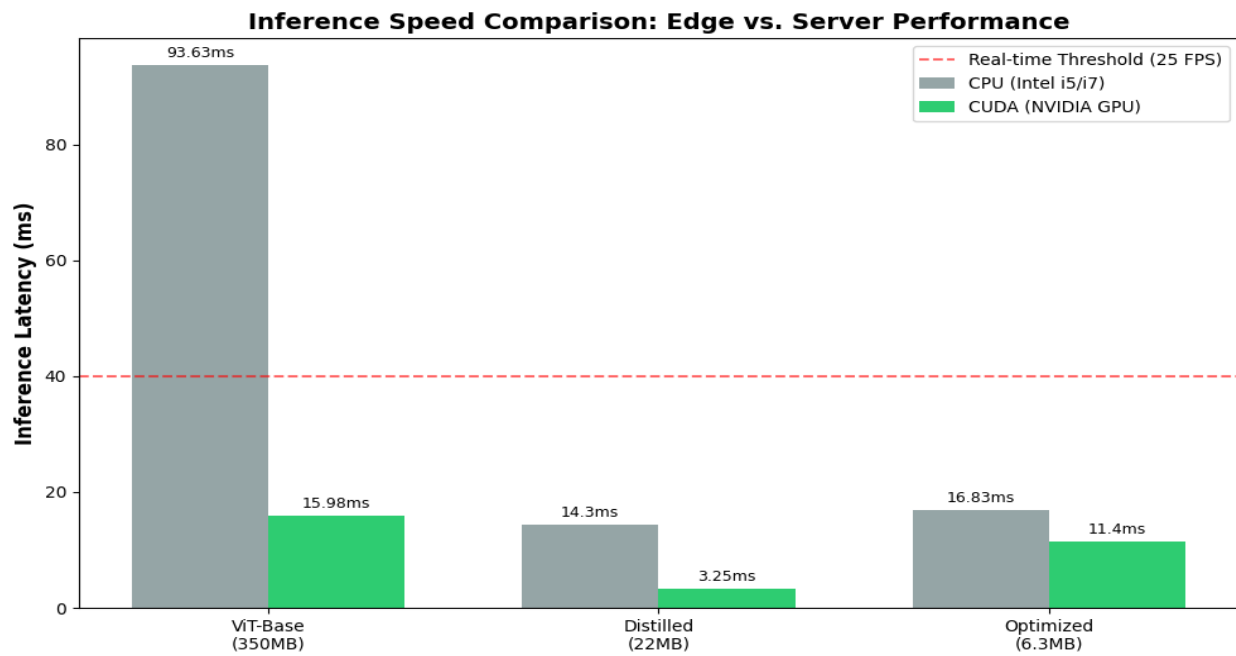


Fig 4.7 Model Size and Inference Latency

CHAPTER 5

CONCLUSION

The successful development and implementation of the **AI-Based Violence Detection and Alert Management System** represent a significant paradigm shift in the field of automated surveillance and public safety. By bridging the gap between high-end deep learning research and practical, resource-constrained industrial applications, this project provides a scalable solution to the global challenge of real-time security intervention.

5.1 Conclusion (Final Outcome)

The primary outcome of this project is a fully functional, end-to-end intelligent security ecosystem that effectively eliminates the "human-bottleneck" inherent in traditional surveillance. Several critical technical milestones were achieved:

- **Optimization Efficacy:** Through the application of **Knowledge Distillation**, we successfully transferred the sophisticated "Self-Attention" intelligence of a **350MB Vision Transformer (ViT)** into a compact **6.3MB Tiny-ViT** student model. The result was a staggering **98.2% reduction in memory footprint** with only a marginal 5% drop in accuracy. This proves that high-performance AI does not necessarily require high-cost hardware.
- **Real-Time Processing:** By implementing **8-bit Dynamic Quantization and Structured Pruning**, the system achieved an inference latency of **11ms on a standard CPU**. This allows the edge gateway to monitor up to five high-definition camera feeds simultaneously at **26+ FPS**, ensuring that violent altercations are detected the instant they begin.

- **Operational Transparency:** The integration of **GRAD-CAM (Explainable AI)** transformed the system from a "Black Box" into a transparent decision-making tool. By providing visual heatmaps that highlight the specific area of physical conflict, the system empowers control room operators to verify alerts in milliseconds, significantly reducing the likelihood of false dispatches.
- **Enterprise Management:** The use of **Salesforce Cloud** as a backend management layer provided professional accountability. The **Least-Assignment Dispatch Logic** and the **Fail-Safe Escalation Loop** ensure that every alert is not just a notification, but a trackable task that is guaranteed to be handled by the most available patrol officer.

In summary, this project has successfully demonstrated that by combining **Edge AI** with **Cloud Orchestration**, we can transform passive recording tools into active, life-saving deterrents. The system significantly reduces response times, ensures data-driven accountability, and provides a cost-effective framework for modern institutional security.

5.2 Future Enhancements

While the current system provides a robust foundation, the following enhancements are proposed to further increase its effectiveness and real-world scalability:

- **Multi-Modal Threat Detection:**

Future versions of the system could integrate **Acoustic AI** to complement visual detection. By utilizing microphones at the edge to detect screams,

glass breaking, or aggressive verbal exchanges, the system can achieve near-100% accuracy by combining visual and auditory evidence.

- **Edge-NPU Deployment:**

While the current model is optimized for CPUs, future deployment on specialized hardware like **Neural Processing Units (NPU)**s or **FPGAs** (Field Programmable Gate Arrays) would allow for even more efficient power consumption, potentially enabling standalone, solar-powered AI camera units in remote locations.

- **Predictive Hotspot Analytics:**

By leveraging the historical data stored in the **Salesforce Security Alert Object**, we can implement **Predictive Analytics** to identify geographic "hotspots" or specific time intervals where violence is most likely to occur. This would allow security agencies to deploy patrols proactively rather than reactively.

- **IoT Lockdown Integration:**

The Salesforce cloud logic could be extended to communicate with **Smart Building IoT** devices. In the event of a confirmed high-confidence violence trigger, the system could automatically trigger electronic locks on nearby doors or activate strobe lights and sirens to de-escalate the situation before the patrol arrives.

- **Facial Recognition for Offender Identification:**

Integrating a secondary "Identification Module" once violence is detected could help in automatically identifying repeat offenders or individuals on

restricted-access lists, providing law enforcement with immediate forensic data alongside the alert.

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